Driver Behavior Due to Cognitive Function Decline of Mild Cognitive Impairment Drivers

軽度認知障害ドライバの認知機能低下による運転行動特性に関する研究

25M11687 Renya Konishi　Supervising Academic Staff : Prof. M. Shino, Assoc.Prof. H. Yoshitake

1.Background

Social Background

Traffic accidents involving older drivers is a social problem in Japan. One of the cause ➔ cognitive decline

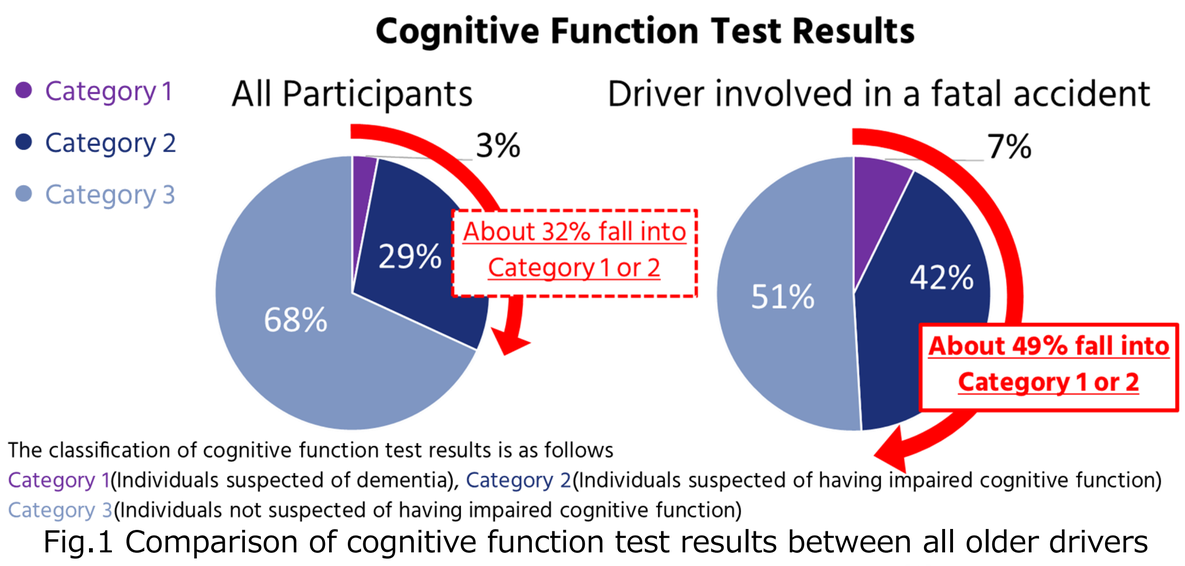


Fig.1 Comparison of cognitive function test results between all older drivers and older drivers involved in fatal accidents[1]

■ **Mild Cognitive Impairment (MCI)**

Cognitive impairment in MCI is **atypical**[5]

Stopping driving increases the risk of dementia
➔ **Continuation of driving** is ideal
Driving Assistance (Support Cars, ADAS, etc.)

Extending the safe driving lifespan of MCI drivers through driving assistance

Research Background

■ Driving behavior characteristics of MCI drivers[5]
➔Not MCI-specific

■ Comparison of driving behavior based on thorough evaluation[6]
➔Link between cognitive impairment and driving behavior remains elusive

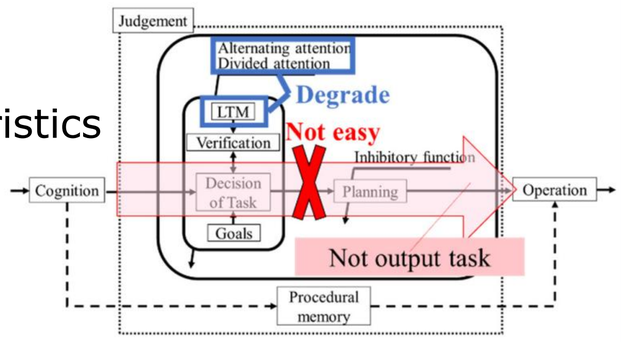


Fig.2 The planning process for older drivers who had dangerous driving behaviors

Linking MCI cognitive impairment to driving behavior for targeted assistance

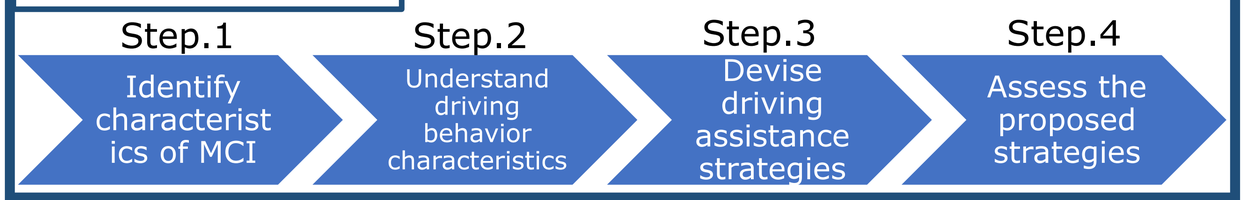
[1]National Police Agency Traffic Bureau. Characteristics of Traffic Fatalities in 2017. 2018[2] Petersen, R.C. and Morris, J.C..Mild cognitive impairment as a clinical entry and treatment target.Arch Neurol.2005,Vol.62,No.7,p. [3] Motoki Shino, Michinobu Nakanishi, Reo Imai, Hiroshi Yoshitake, Yoshio Fujita.Investigation of Driving Behavior and Cognitive Ability Concerning Planning Process during Driving of Elderly Drivers.International Journal of Automotive Engineering.2018,Vol.9,No.3,pp.138-144.[4] Pavlou D., Papadimitriou E., Antoniou C., Papantoniou P., Yannis G., Golias J., Papageorgiou S.G..Driving Behaviour of Drivers with Mild Cognitive Impairment and Alzheimer's Disease: A Driving Simulator Study.TRB.2015

2.Objective & Research Flow

Objective

Propose driving assistance considering driving behavior characteristics due to cognitive impairment in MCI drivers

Research Flow



3.Research Progress

Step.1 Identify characteristics of MCI

Decline in memory function[5]
◆ Decreased learning speed
◆ Decline in retrieving information

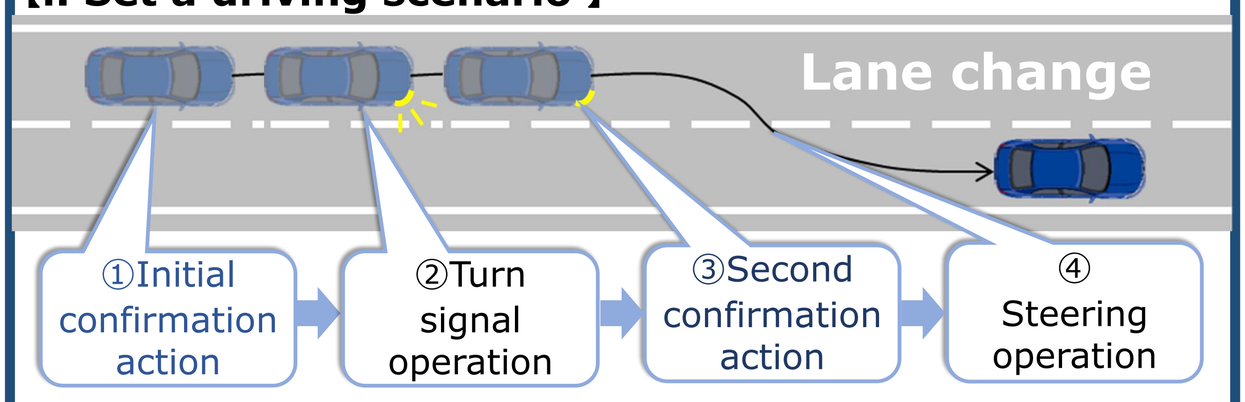
Impairment in iADL (instrumental activities of daily living)[6]

Instability in finger tapping movements[7]

[5] Kaarin J. Ansteya, Ranmalee Eramudugollaa, Sidhant Chopraa, Jasmine Pricea and Joanne M. Woodb.Assessment of Driving Safety in Older Adults with Mild Cognitive Impairment.Journal of Alzheimer's Disease.2017,57,1197-1205.[6] ROBERT PERNECZKY, CORINA POHL, CHRISTIAN SORG, JULIA HARTMANN, KATJA KOMOSSA,PANAGIOTIS ALEXOPOULOS, STEFAN WAGENPFEIL, ALEXANDER KURZ.Complex activities of daily living in mild cognitive impairment: conceptual and diagnostic issues.Age and Ageing.2006,35,240-245.[7] Yuko Sano, Shota Suzumura, Junpei Sugioka, Tomohiko Mizuguchi, Akihiko Kandori, Izumi Kondo.Detecting mild cognitive impairment by applying integrated random forest to finger tapping.Medical & Biological Engineering & Computing.2025,63,1881-1894.

Step.2 Understand driving behavior characteristics in MCI drivers

【i. Set a driving scenario 】



【ii. Formulate hypothesis】

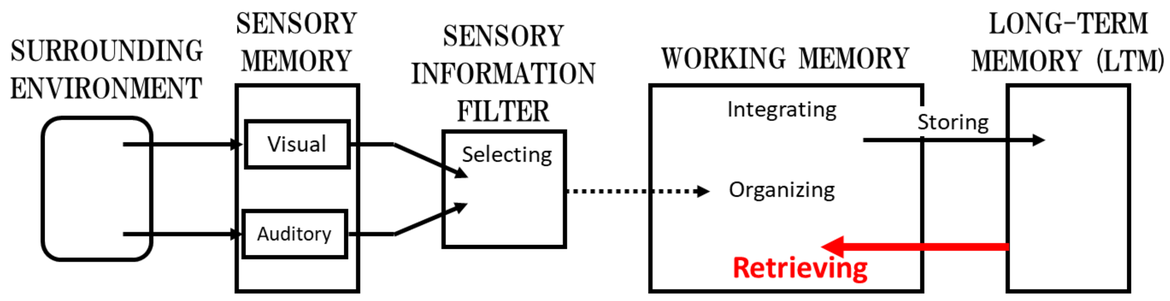


Fig.3 Information processing model[8]

Performed ① ➔ Information about performing confirmation action exists within LTM

Omitted ③ ➔ The information exists in LTM but failed **retrieving**

Hypothesis: MCI drivers are prone to omitting the second confirmation action

【iii. Verify hypothesis】

■ Experimental data used for the analysis

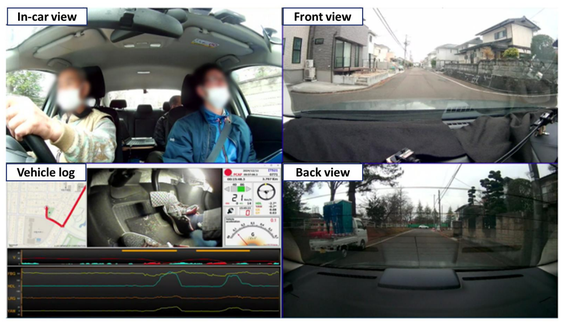


Fig.4 Driving data

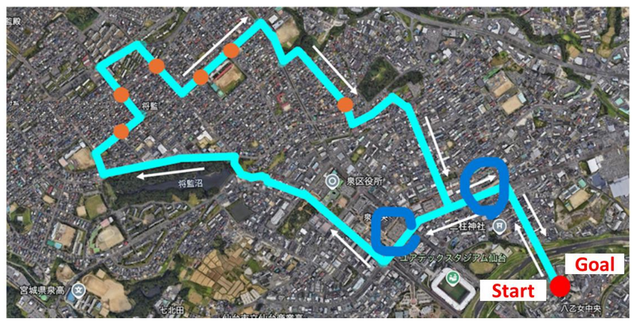


Fig.5 Driving course

◆ Participants
Older drivers aged 65 and over (n=21)

◆ Analysis subject
Data from **14** participants who performed the initial confirmation action when changing lanes

■ Classifying MCI/non-MCI
MCI if
◆ MMSE score＜24
◆ WMS-RⅡ score＜8

Result (Fig.6)
MCI(6) / non-MCI(8)

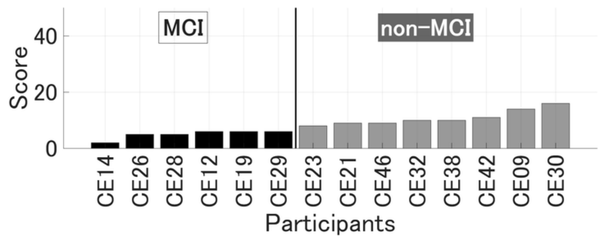


Fig.6 MCI/non-MCI classification results based on WMS-RⅡ score

■ Driving behavior analysis
Result (Fig.7)
MCI: **50**% > non-MCI: **38**%
Omitted(second confirmation action)

No significant difference
$p = 0.529$ (>0.05)

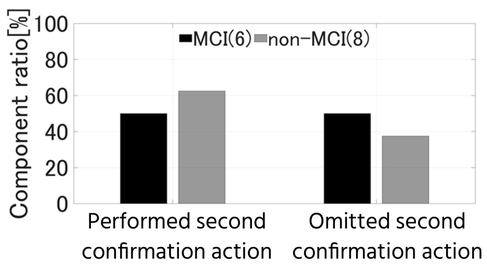


Fig.7 Comparing ratio (second confirmation action) in MCI and non-MCI group

[8] Moreno, R. and Mayer, R. (2007). Interactive multimodal learning environments. Educational Psychology Review, 19:309–326.

4.Summary & Future plans

We formulated a hypothesis regarding MCI drivers' driving behavior. The experimental results showed a tendency to support this hypothesis.

Future plan
- ☐ Examine relationship between driving behavior and cognitive function in further detail(Step.2)
- ☐ Devise driving assistance strategies(Step.3)
- ☐ Conduct experiments to evaluate the strategies(Step.4)

Assistive Technology Lab., Dept. of Mechanical Engineering, School of Engineering, Institute of Science Tokyo